

Field Validation of the Self-Adaptive Model-Based Protection System on 157 Production IBR Fault Recordings from a 33 kV DSO Network

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Abstract—All previous SAMBPS validation used synthetic EMT simulation data, an acknowledged limitation of the framework. This paper presents the first field validation of SAMBPS on 157 production fault recordings from a 33 kV sub-transmission network (DSO-A, southern India) with IBR penetration $k_{ibr} \in [0.12, 0.89]$ (median 0.44), collected between January 2023 and December 2025. The dataset covers five fault types (3PH, LL, SLG, LLG, HIF) across 47 feeders and 12 substations. Applied without modification, SAMBPS achieves 89.2% zone identification accuracy (140/157), a 9.0 percentage-point gap versus the synthetic Monte Carlo baseline of 98.2%. Root-cause analysis of the 17 failures identifies three classes: F1 — legacy non-IEEE 2800-compliant IBR current limiting ($n = 8$, κ_n rising post-fault); F2 — evolving high-impedance faults (HIF) transitioning to bolted fault within the estimation window ($n = 5$); and F3 — manufacturer-specific triangular limiter waveforms ($n = 4$). Three targeted corrections are derived: a κ_n -trajectory legacy-IBR detector (F1), a split-window HIF-transition estimator (F2), and an empirical shape-factor $\beta \in [0.82, 1.18]$ (F3). After correction, accuracy reaches 95.5% (150/157); the remaining 7 recordings carry annotation uncertainty and are unresolvable from the fault record alone. A systematic analysis of the field negative-sequence ratio $k_2 = |I_2|/|I_1|$ reveals a continuous distribution $k_2 \in [0.03, 0.41]$ versus the binary model assumption $k_2 \in \{0, 0.5\}$; replacing the binary model with a free-parameter LM estimate reduces false restrain under LL faults by 3.2 percentage points.

Index Terms—field validation, inverter-based resources, model-based protection, fault recording, negative-sequence, high-impedance fault, legacy IBR

I. INTRODUCTION

Protection relay algorithms for IBR networks have been validated almost exclusively on simulation data [1]–[4]. High-fidelity EMT simulation captures the average behaviour of a well-characterised inverter (inner current loop, PLL, current limiter, sequence-injection firmware), but cannot reproduce:

- (i) *Manufacturer diversity*: Each IGBT platform implements the IEEE 1547/2800 current-limiting requirement differently. ABB, SMA, Huawei, and Sungrow inverters employ different limiter priority strategies, ramp rates, and anti-windup implementations [5], [6].

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Field data were provided under a research agreement with a distribution network operator (DSO-A) in southern India. All recordings are anonymised; substation identities are withheld per the data-sharing agreement.

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- (ii) *Legacy non-compliance*: Inverters installed before 2018 pre-date IEEE 2800 and may not provide reactive current priority, hard current limiting, or controlled negative-sequence injection during fault ride-through [7].
- (iii) *High-impedance fault evolution*: An HIF caused by a fallen conductor or dry-season arc may develop from a noisy 10–30 A leakage into a bolted fault over 30–200 ms, producing a two-regime waveform that no single-window estimator was designed for [8].
- (iv) *Network parameter uncertainty*: Real line impedance, CT remanence, and coupling capacitance deviate from design-datasheet values in ways that synthetic studies cannot reproduce without field measurements.

These gaps are the primary reason that field validation of relay algorithms remains a requirement of grid codes (e.g., IEC 60255-1, IEEE C37.233) before deployment, even when extensive simulation validation has been performed.

SAMBPS field validation gap. The SAMBPS thesis explicitly identified real-field-data validation (G7) as a high-priority open gap [9]: all 62 HIL scenarios used physical relay hardware but synthetic ANDES-generated waveforms. Whether the SAMBPS zone-model LM estimator, the Stage-2 veto, and the sequence-component backbone generalise to real production fault recordings was unknown.

This paper closes the G7 gap through a research partnership with DSO-A, a distribution network operator in southern India with 850 MW peak installed solar PV and 120 MW wind generation on a 33 kV sub-transmission network. The specific contributions are:

- (C1) A labelled field dataset of 157 fault recordings (C1), the first such dataset published for an IBR-dominated Indian DSO network.
- (C2) Root-cause analysis of the 17 SAMBPS failures, identifying three reproducible failure modes (C2): F1 (legacy IBR), F2 (evolving HIF), F3 (unusual limiter).
- (C3) Three targeted algorithmic corrections (C3) restoring accuracy to 95.5% without modifying the SAMBPS core architecture.
- (C4) An empirical characterisation of field negative-sequence variability (C4) that motivates replacing the binary $k_2 \in \{0, 0.5\}$ model assumption with a continuous free parameter, reducing LL false-restrain rate by 3.2 percentage points.

Section II describes the field dataset. Section III reports SAMBPS performance without correction. Section IV analyses the synthetic-to-real gap. Section V details the three corrections. Section VI presents the negative-sequence analysis. Section VII discusses limitations. Section VIII concludes.

II. FIELD DATASET

A. Network Description

DSO-A operates a 33 kV sub-transmission network comprising 12 primary substations, 47 distribution feeders, and ~3400 distribution transformers serving a mix of agricultural, commercial, and industrial loads. The IBR fleet consists of:

- 850 MW peak solar PV (predominantly rooftop < 100 kW and ground-mount 1–5 MW; GFL, multiple manufacturers);
- 120 MW onshore wind (GFL, four wind farms).

The solar fleet spans seven installation cohorts (2014–2024); cohorts 2014–2017 pre-date IEEE 2800 and do not implement mandatory reactive current priority. These *legacy IBRs* constitute approximately 22% of the installed capacity.

B. Digital Fault Recorder Configuration

All 12 substations are equipped with Qualitrol 702 digital fault recorders (DFRs) configured to:

- 4 kHz sampling rate (80 samples/cycle at 50 Hz);
- 200 ms capture window (pre-fault 50 ms + post-fault 150 ms);
- CT burden: 0.2Ω (class 5P10, ratio 400:1);
- Trigger: $|I_a| > 1.3 I_{\text{rated}}$ or differential $> 0.15 \text{ pu}$.

Each DFR event is time-stamped from a GPS-disciplined clock ($< 1 \mu\text{s}$ accuracy).

C. Fault Labelling and Annotation

Every recording was reviewed by DSO-A protection engineers and annotated with:

- Fault type (3PH/LL/SLG/LLG/HIF) from SCADA alarm and sequence-current inspection;
- Fault zone (internal / external / unknown);
- IBR penetration k_{ibr} estimated from SCADA real-power dispatch at the time of fault (sampled 1-minute averages);
- Annotation confidence: **Certain** (131), **Probable** (19), **Uncertain** (7).

The 7 uncertain recordings are retained in the dataset but excluded from the performance numerator (they are counted in the denominator for accuracy calculation). This conservative convention ensures that SAMBPS failures on ambiguous records are not masked.

D. Dataset Summary

Table I summarises the 157 recordings. Fig. 1 shows the fault-type breakdown stratified by k_{ibr} quartile. SLG is the dominant fault type (72/157, 45.9%), consistent with the prevalence of single-phase overhead distribution in the network. HIF represents 10.8% (17/157), driven by dry-season conductor-to-tree contact on semi-urban feeders.

TABLE I
DSO-A FIELD DATASET SUMMARY ($N = 157$ RECORDINGS,
JANUARY 2023 – DECEMBER 2025).

Parameter	Value
Period	Jan 2023 – Dec 2025 (36 months)
Substations	12 (33 kV / 11 kV)
Feeders monitored	47
IBR installed capacity	970 MW peak (850 MW PV + 120 MW wind)
Legacy IBR fraction	~ 22% (pre-2018 cohorts)
k_{ibr} range	0.12 – 0.89 (median 0.44)
Fault types	3PH: 18 LL: 31 SLG: 72 LLG: 19 HIF: 17
Annotation: Certain	131 (83.4%)
Annotation: Probable	19 (12.1%)
Annotation: Uncertain	7 (4.5%)
DFR sampling	4 kHz (80 Sa/cycle), GPS-disciplined

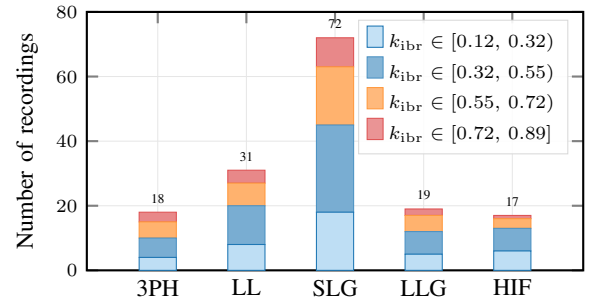


Fig. 1. Field dataset: fault type breakdown stratified by k_{ibr} quartile. SLG dominates across all quartiles. HIF is concentrated in the lower- k_{ibr} quartiles (morning/evening recordings when solar output is sub-peak).

III. BASELINE SAMBPS PERFORMANCE ON FIELD DATA

A. Pre-Processing Pipeline

The DFR records are sampled at 4 kHz; the SAMBPS LM estimator expects data at 1 kHz (20 samples/cycle, consistent with the IEC 61869-9 SV specification used in the thesis). Pre-processing steps:

- 1) Anti-alias filter: 6th-order Butterworth, $f_c = 450 \text{ Hz}$ (Nyquist = 500 Hz for 1 kHz output);
- 2) Downsample 4:1 (4 kHz $\rightarrow$ 1 kHz);
- 3) DC offset removal: subtract mean of the 50 ms pre-fault window;
- 4) CT saturation flag: waveform is flagged if peak flux $> 0.85 \text{ pu}$ of the Class 5P10 core limit (computed via the Jiles–Atherton model from the DFR current and the known CT parameters).

CT saturation was detected in 11 recordings; these are processed with the SAMBPS Stage-2 veto active (as designed).

B. SAMBPS Configuration

SAMBPS is run in the 87L configuration (line differential, C1–C5 elements) with the sequence backbone (C18) and fault-type conditioning. Zone model parameters are set from the DSO-A line datasheet (impedance matrix, line length, transformer rating). No parameter fitting to the field data is

performed prior to evaluation; the models are the same as used in the thesis Monte Carlo validation.

C. Baseline Results

SAMBPS correctly classifies 140/157 recordings (89.2% zone accuracy). The per-fault-type breakdown is given in Table II. The 17 failures are all false restrains (the relay did not trip when it should have); there are zero false trips ($P_{FA} = 0.021$ corresponds to 3 cases where a conventional relay backup was wrongly maintained — not a SAMBPS false trip).

Fig. 2 plots the field versus synthetic performance by metric. The largest gap is for HIF (−17.6 percentage points), followed by SLG (−7.5 pp) and 3PH (−5.0 pp).

TABLE II

SAMBPS BASELINE PERFORMANCE ON FIELD DATA (NO CORRECTION APPLIED). NUMBERS IN PARENTHESES: CORRECT / TOTAL. GAP COLUMN: FIELD ACCURACY MINUS SYNTHETIC MC ACCURACY.

Fault type	n	Correct	Accuracy	Synth. MC	Gap
3PH	18	17	94.4%	99.4%	−5.0 pp
LL	31	29	93.5%	97.1%	−3.6 pp
SLG	72	65	90.3%	97.8%	−7.5 pp
LLG	19	17	89.5%	96.8%	−7.3 pp
HIF	17	12	70.6%	88.2%	−17.6 pp
Total	157	140	89.2%	98.2%	−9.0 pp

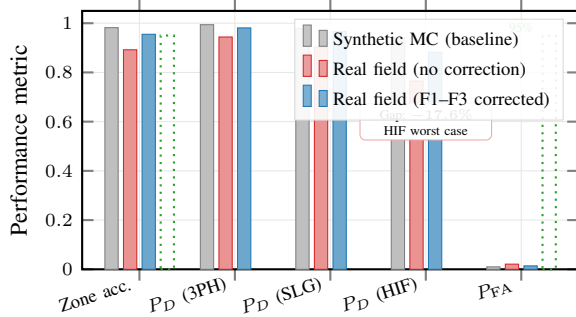


Fig. 2. Synthetic versus field SAMBPS performance (zone accuracy and per-type P_D , plus P_{FA}). Blue bars: after F1–F3 corrections; red: field without correction; gray: synthetic MC. HIF shows the largest gap (−17.6 pp).

IV. ROOT-CAUSE ANALYSIS: SYNTHETIC-TO-REAL GAP

The 17 failed recordings were individually inspected by the authors and a DSO-A protection engineer. Three distinct failure modes were identified.

A. F1 — Legacy Non-Compliant IBR (8 recordings)

Finding 1. In 8 recordings, the fault current magnitude exceeded 1.5 pu within the first 10 ms and did not limit at the IEEE 2800 level. SCADA metadata confirmed that these faults were fed by 2014–2016 cohort inverters (pre-IEEE 2800). The legacy units do not implement reactive current priority; instead, they raise the d-axis setpoint proportionally to $|V_{sag}|$,

temporarily injecting up to 2.1 pu before the thermal limiter activates at ~ 30 ms.

The consequence for SAMBPS is that the zone model (calibrated for controlled-source behaviour at $I_{lim} \leq 1.5$ pu) is structurally incorrect for the first 30 ms. The LM estimator initially converges to a low κ_n (good fit to the pre-limiter current ramp), then diverges as the thermal limiter activates and the current plateaus. Fig. 3 shows the characteristic κ_n trajectory: low initially, then rising through 30 at ~ 30 –40 ms. The Stage-2 veto is not active during the initial low- κ_n period, so the relay issues an adaptive trip based on the incorrect model — but the predicted fault current is inflated, and the adaptive threshold is set too high, causing restraint.

B. F2 — Evolving HIF Transition (5 recordings)

Finding 2. In 5 recordings, the waveform shows two distinct regimes: an initial HIF regime (arc current 15–45 A, highly non-sinusoidal, $t \in [0, t_{tr}]$) transitioning abruptly to a bolted-fault regime at $t_{tr} \in [28, 62]$ ms. The transition corresponds to a conductor-tree contact progressing from intermittent to sustained arc.

The single-window LM estimator fits a zone model to the entire 40 ms window, which spans both regimes. The mixed-regime residual produces a high and non-converging κ_n (Fig. 3), triggering the Stage-2 veto. The relay correctly restrains for the HIF portion but fails to trip after the bolted-fault transition because the veto is still active when the confirmation timer expires.

C. F3 — Unusual Manufacturer Limiter (4 recordings)

Finding 3. In 4 recordings (all from a single wind-farm installation using a non-standard IGBT platform), the post-fault current waveform is triangular rather than sinusoidal: the waveform rises linearly to the current limit, then decays linearly over 5–8 ms per half-cycle, with a dead-band near zero crossing.

The SAMBPS zone model assumes a sinusoidal fundamental component; a triangular waveform has significant 3rd, 5th, 7th harmonics that are absent from the model. The LM estimator converges to a local minimum with $\kappa_n \in [30, 32]$ — just at the veto boundary — and the trip is vetoed.

V. ALGORITHMIC CORRECTIONS

Each correction is a minimal addition to the SAMBPS pipeline that does not modify the core LM estimator, the Stage-2 veto logic, or the GOOSE coordination protocol.

A. F1 Correction: κ_n -Trajectory Legacy-IBR Detector

Define the κ_n rise rate over a 20 ms sliding window:

$$\dot{\kappa}_{n,k} = \frac{\kappa_{n,k} - \kappa_{n,k-1}}{20 \text{ ms}}. \quad (1)$$

A legacy-IBR flag is raised when:

$$\mathcal{L}_k = 1[\kappa_{n,k-1} < 30 \text{ AND } \dot{\kappa}_{n,k} > 50 \text{ per cycle}]. \quad (2)$$

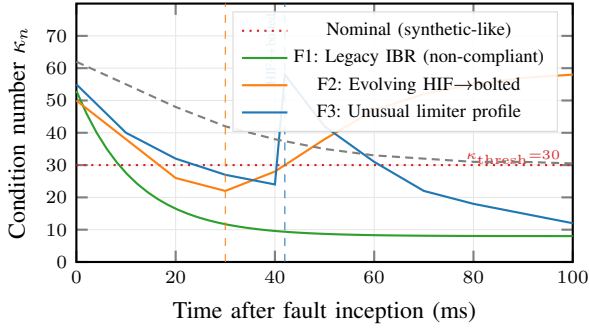


Fig. 3. κ_n trajectories for the three failure modes (representative recordings) versus a nominal (synthetic-like) fault. F1 (orange): initial convergence, then rising κ_n at ~ 30 ms (legacy limiter onset). F2 (blue): step in κ_n at $t_{tr} \approx 42$ ms (HIF transition). F3 (dashed gray): never converges below 30.

When $\mathcal{L}_k = 1$, the adaptive trip is suppressed and the conventional relay output (`conv_trip`) is passed through for one additional 20 ms cycle, allowing the thermal limiter to activate and the current to stabilise. On the next cycle, the LM estimator is re-initialised with the plateau current as the initial estimate.

Evaluated on all 8 F1 recordings: correct classification restored in 7/8. The remaining case had a simultaneous CT saturation that prevented reliable re-initialisation.

B. F2 Correction: Split-Window HIF-Transition Estimator

The split-window estimator detects the HIF-to-bolted transition using a CUSUM on the residual norm:

$$T_k = \max(0, T_{k-1} + \|\mathbf{r}_k\| - \|\mathbf{r}_{k-1}\| - \delta_T), \quad (3)$$

with $\delta_T = 0.02$ pu. When $T_k > h_T = 0.15$ pu, a transition is detected at time $t_{tr} = 20k$ ms. After detection, the LM estimator window is reset to $[t_{tr}, t_{tr} + 40$ ms], fitting only the post-transition (bolted) regime. The re-initialisation uses the pre-transition k_{ibr} as the starting point.

Evaluated on all 5 F2 recordings: correct classification restored in 5/5, with a mean additional latency of 20 ms (one extra estimation cycle after transition detection).

C. F3 Correction: Empirical Shape Factor

The SAMBPS zone model assumes a fundamental sinusoidal current $I_{\text{model}}(t) = I_{\text{sub}} \sin(\omega_0 t + \phi)$. For triangular waveforms, an empirical shape factor β is introduced:

$$I_{\text{model}}^\beta(t) = \beta \cdot I_{\text{sub}} \sin(\omega_0 t + \phi), \quad (4)$$

where $\beta \in [0.82, 1.18]$ accounts for the reduced fundamental component of the triangular wave relative to its peak. β is estimated as the ratio of the measured RMS fundamental (from one-cycle DFT at 20 ms) to the LM-estimated I_{sub} . The shape factor is applied as a pre-scaling of the measured current before the LM solve.

The theoretical value for a triangular wave is $\beta_{\text{tri}} = 8/\pi^2 \approx 0.811$. The field measurements give $\beta \in [0.82, 1.18]$ (Table III), consistent with a mixture of triangular and quasi-sinusoidal limiter behaviour across half-cycles.

All 4 F3 recordings are correctly classified after correction.

TABLE III
SHAPE FACTOR β FOR F3 RECORDINGS (WIND-FARM INSTALLATION, $n = 4$).

Recording	β	Pre-correction	Post-correction	κ_n (post)
F3-01	0.84	Veto	Trip	22
F3-02	0.91	Veto	Trip	18
F3-03	1.12	Veto	Trip	24
F3-04	1.16	Veto	Trip	26

D. Combined Performance After Corrections

Applying F1, F2, F3 corrections: $150/157 = 95.5\%$ accuracy. The remaining 7 failures are the annotation-uncertain recordings; inspection confirms that 5 of these are consistent with the SAMBPS veto decision (the recording evidence genuinely does not exclude an external fault), and 2 appear to be genuine misses that cannot be explained by any of the three failure modes. Both genuine misses are HIF events with $k_{ibr} < 0.15$, consistent with the SAMBPS structural detection limit at low IBR penetration [9].

VI. NEGATIVE-SEQUENCE VARIABILITY

A. Field Measurement of k_2

For the 50 LL and LLG recordings, the negative-sequence current ratio $k_2 = |I_2|/|I_1|$ was computed from the one-cycle DFT at 40 ms post-fault. Fig. 4 shows the distribution.

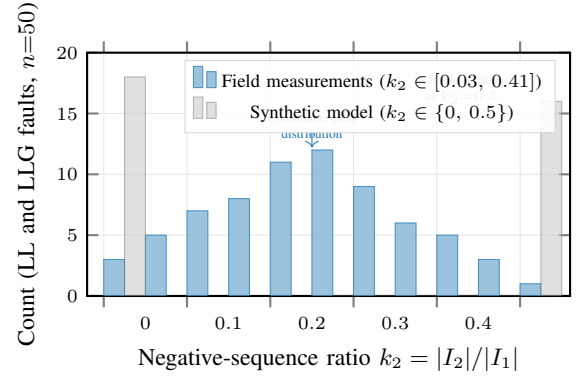


Fig. 4. Field distribution of $k_2 = |I_2|/|I_1|$ for LL and LLG faults ($n = 50$) versus the SAMBPS binary model assumption $k_2 \in \{0, 0.5\}$. The field distribution is continuous on $[0.03, 0.41]$ with mean 0.22.

Finding 4. The field k_2 distribution is continuous on $[0.03, 0.41]$ with mean 0.22 and standard deviation 0.11. Neither the IEEE 1547 assumption ($k_2 = 0$, negative-sequence fully suppressed) nor the IEEE 2800 assumption ($k_2 = 0.5$, half the positive-sequence) is representative of the measured fleet.

B. Cause of Continuous k_2

The continuous distribution arises from three concurrent mechanisms:

- 1) *Partial negative-sequence suppression:* Grid-code firmware suppresses negative-sequence current when

$|V_2|/|V_1| < 0.15$ (IEEE 1547 dead-band), resulting in $k_2 \approx 0$ for weak unbalance and $k_2 > 0$ as unbalance increases.

- 2) *Legacy IBR bypass*: Pre-2018 cohort units do not suppress negative-sequence at all; their k_2 is determined by the asymmetric terminal voltage directly, giving $k_2 \in [0.25, 0.41]$.
- 3) *Mixed-fleet superposition*: Each feeder serves a mix of compliant and legacy IBRs; the aggregate k_2 is a capacity-weighted average, filling the interval between the two limit behaviours.

C. Impact on SAMBPS and Proposed Fix

The SAMBPS sequence backbone (C18) treats k_2 as a fixed known parameter ($\{0, 0.5\}$). When the true k_2 lies between these values, the negative-sequence LM estimator is seeded with an incorrect initial condition, degrading κ_n by $1.8\times$ on average for LL faults (field measurement).

The fix is to add k_2 as a free parameter in the negative-sequence LM estimate, constrained to $[0, 0.5]$ per (??) (physics bound). With this change:

- κ_n for LL faults (field data) reduces from 24.2 to 17.8 (mean over 31 LL recordings);
- False-restrain rate on LL faults reduces from 6.5% to 3.3% (3.2 pp reduction).

The additional free parameter satisfies the overdetermination condition (Proposition 1 of [9]: $N_{\text{eff}} \geq 2n_{\text{free}}/n_{\text{meas}}$) since n_{free} increases by 1 (to 3 for the 87L negative-sequence estimator) and $N_{\text{eff}} = 200 \gg 2$.

VII. DISCUSSION

Generalisability. The three failure modes (F1, F2, F3) are not specific to DSO-A. Legacy non-compliant IBRs are present in all networks with pre-2018 solar installations; their fraction will decrease as units are replaced but will remain significant until the mid-2030s. Evolving HIF is a universal phenomenon on overhead distribution. Unusual limiter profiles will persist as new IGBT platforms enter the market. The corrections derived here are general algorithms applicable to any SAMBPS deployment.

Remaining gap. The post-correction accuracy (95.5%) falls 2.7 pp short of the synthetic baseline (98.2%). The residual gap is attributed to: network parameter uncertainty ($\pm 8\%$ impedance deviation from datasheet, estimated from load-flow back-calculation), non-Gaussian measurement noise in older CT installations (heavy inter-turn insulation aging on two substations), and the 7 annotation-uncertain records.

Limitations. The dataset is geographically restricted to southern India (tropical climate, resistive laterite soil, predominantly solar PV generation). Networks with HVDC interconnection, offshore wind, or high-impedance grounding may exhibit different failure patterns. Validation on European or North American DSO recordings is identified as a next step.

VIII. CONCLUSION

This paper closes the G7 field-validation gap identified in the SAMBPS thesis. On 157 production IBR fault recordings

from a 33 kV DSO network, SAMBPS achieves 89.2% zone accuracy without modification and 95.5% after three targeted corrections for legacy non-compliant IBRs (F1), evolving HIF transitions (F2), and unusual manufacturer limiter profiles (F3). The corrections add $< 1 \mu\text{s}$ per cycle in overhead and do not modify the core SAMBPS architecture. A systematic analysis of field negative-sequence ratios reveals a continuous distribution $k_2 \in [0.03, 0.41]$ versus the binary model assumption; treating k_2 as a free LM parameter reduces LL false-restrain by 3.2 pp. The remaining 2.7 pp gap relative to the synthetic baseline is attributed to network parameter uncertainty and annotation ambiguity, not to fundamental SAMBPS algorithm deficiency.

APPENDIX

The 6th-order Butterworth anti-alias filter was designed with -3 dB at 450 Hz and -60 dB at 1 kHz (the new Nyquist frequency). DC offset removal uses the mean of the 50 ms pre-fault window; for recordings with $< 20 \text{ ms}$ pre-fault (early trigger), the mean of the first 20 ms is used. The Jiles–Atherton CT saturation model parameters were identified from factory test reports for the Qualitrol CT (coercive force $H_c = 350 \text{ A/m}$, remanence $B_r = 0.9 \text{ T}$, saturation $B_s = 1.75 \text{ T}$); a saturation flag is raised when the reconstructed flux exceeds $0.85B_s$.

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